Automated Multi-Label ICD Coding from Clinical Notes

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1. Introduction

In modern healthcare, clinical documentation serves as the primary record of patient care, yet its transformation into actionable administrative data remains a significant bottleneck. International Classification of Diseases (ICD) coding is the process of assigning standardized alphanumeric codes to unstructured clinical notes — such as discharge summaries and operative reports — to facilitate billing, epidemiological tracking, and clinical research. Traditionally, this task is performed manually by trained clinical coders who interpret complex narratives to identify relevant diagnoses and procedures. However, the exponential growth of medical data and the increasing granularity of coding systems have led to manual coding becoming increasingly inefficient; the transition from ICD-9 to ICD-10 expanded the code set from approximately 14,000 to over 70,000 labels, substantially increasing the cognitive burden on human coders (Motzfeldt et al., 2025).

The emergence of Natural Language Processing (NLP) and Machine Learning (ML) offers a promising path toward automating this process. By leveraging large-scale datasets like the Medical Information Mart for Intensive Care (MIMIC-IV), researchers can develop models capable of reading clinical text and predicting appropriate ICD labels (Edin et al., 2023; Johnson et al., 2016). This paper presents and evaluates an automated multi-label classification system focused on the top 50 most frequent ICD-10 codes from the MIMIC-IV clinical database, comparing a traditional statistical baseline against a fine-tuned transformer-based architecture.

Importantly, this study does not assume that deep learning models will automatically outperform simpler baselines. The results reported here contribute to an emerging body of evidence showing that truncation constraints can critically impair transformer models on long clinical documents, leading to cases where TF-IDF-based methods remain competitive or superior (Edin et al., 2023).

2. Background and Motivation

2.1 Financial and Operational Impact

The motivation for automating medical coding is driven by immense economic pressure and operational complexity. Inaccurate coding is estimated to cost the U.S. healthcare industry approximately $36 billion annually due to lost revenue, denied insurance claims, and administrative penalties (American Medical Association, 2024; Crowe RCA, 2022). Industry benchmarks indicate that roughly 11% of all claims are denied, which for an average-sized health system can equate to over 110,000 unpaid claims per year. This results in cumulative revenue leakage often reaching 10–15% of a practice's annual earnings (BlueBrix, 2022). Table 1 summarizes the key financial metrics driving the need for automation.

Table 1. Financial and operational impact of medical coding errors.

Metric	Value / Source
Industry annual cost of coding errors	$36 billion (AMA, 2024)
Average claim denial rate	11% (Crowe RCA, 2022)
Revenue leakage per practice	10–15% of annual earnings (BlueBrix, 2022)
Avg coding time per inpatient record (ICD-10)	~40 minutes (Edin et al., 2023)
CMS/OIG fines (2019)	$6.2 billion combined (AMA, 2024)

2.2 Complexity of the ICD-10 Transition

The transition from ICD-9 to ICD-10 substantially increased cognitive load on human coders. ICD-10 provides more detailed disease classification, expanding the available code set to over 70,000 categories in the U.S. Clinical Modification (Motzfeldt et al., 2025). The average time required to code an inpatient record has consequently risen to nearly 40 minutes (Edin et al., 2023). Furthermore, the long-tail distribution of codes — where a small subset of codes appears frequently while thousands appear rarely — creates a significant challenge for both human experts and automated systems (Li et al., 2020; Mullenbach et al., 2018). This study focuses on the top 50 most frequent codes, representing a tractable subset of this problem while covering the diagnoses that account for the largest share of clinical encounters.

3. Literature Review

The evolution of automated ICD coding has moved from simple statistical heuristics to complex neural architectures that approach human-level reasoning on curated benchmarks.

3.1 Traditional Keyword-Based Baselines

Early attempts at automation treated medical coding as a text classification problem. These models relied on Term Frequency-Inverse Document Frequency (TF-IDF) to identify important terms in a clinical note and logistic regression to predict the associated codes (Nielsen & Kılıç, 2025). TF-IDF highlights terms that are unique to a specific document rather than common across all records. While efficient at identifying high-frequency diagnoses, these models are context-blind: they cannot recognize when a medical term is used negatively (e.g., "no history of diabetes") or understand synonyms (Nielsen & Kılıç, 2025; Shamatrin, 2025). Critically, however, TF-IDF operates over the full document without truncation, giving it a potential advantage over token-limited transformer models on very long clinical notes.

3.2 Attention-Based Neural Models

A major breakthrough occurred with the Convolutional Attention for Multi-Label classification (CAML) model, which introduced label-wise attention — allowing the model to attend to different parts of a long clinical note for each code simultaneously (Mullenbach et al., 2018). Subsequently, the Label Attention Model (LAAT) and its extension JointLAAT used Bidirectional Long Short-Term Memory (BiLSTM) networks to read text both forward and backward, capturing long-distance context. By training the model to recognize broader parent categories at the same time as specific codes, JointLAAT improved accuracy for rare conditions with limited training data (Li et al., 2020; Vu et al., 2020).

3.3 Domain-Specific Transformer Models

The current standard in NLP is the BERT transformer architecture. However, a BERT model trained on general text (Wikipedia, books) often fails to understand the specialized shorthand and abbreviations used in clinical documentation (Alsentzer et al., 2019; Huang et al., 2023). To address this, ClinicalBERT and BioClinicalBERT were pre-trained on millions of hospital notes from the MIMIC-III database, enabling better interpretation of medical terminology (Alsentzer et al., 2019). The PLM-ICD model currently represents the state of the art by combining domain-specific embeddings with a chunking method that allows processing of extremely long discharge summaries beyond the 512-token limit (Edin et al., 2023).

Importantly, Edin et al. (2023) demonstrated that naive truncation of long discharge summaries — as used in this study — substantially degrades transformer performance because diagnostically critical information frequently appears in the latter portions of the document. This finding is directly relevant to interpreting the results presented here.

3.4 Agentic AI Workflows

The most recent paradigm shift is the move toward agentic systems such as "Code Like Humans" (CLH) and MedDCR (Motzfeldt et al., 2025; Wang et al., 2025). Rather than making a one-shot prediction, these systems use Large Language Models (LLMs) to follow a professional coding workflow: (1) extracting evidence from clinical text, (2)

mapping mentions to standard ICD terms, (3) validating the code hierarchy, and (4) reconciling conflicting codes. This approach is particularly effective for open-set coding, where the model must assign codes from the full 70,000+ label space including codes never seen during training (Motzfeldt et al., 2025).

4. Research Question and Objectives

4.1 Research Question

How effectively can NLP-based machine learning models predict the top 50 ICD-10 diagnosis codes from clinical discharge summaries, and what are the relative strengths and limitations of traditional TF-IDF baselines compared to fine-tuned transformer-based models under real-world document-length constraints?

4.2 Research Objectives

- Develop and evaluate a TF-IDF + logistic regression baseline for multi-label ICD prediction from clinical discharge summaries.
- Fine-tune a domain-specific transformer model (BioClinicalBERT) on the same task and compare its performance to the baseline.
- Investigate the effect of classification threshold selection on transformer performance through a systematic threshold sweep.
- Analyze error patterns and performance variation across ICD codes of different frequency levels.

5. Methodology

5.1 Dataset Description

This study uses the Medical Information Mart for Intensive Care (MIMIC)-IV v2.2 and MIMIC-IV-Note v2.2 datasets. MIMIC-IV contains comprehensive hospital data for approximately 250,000 patients admitted to the emergency department or intensive care unit at Beth Israel Deaconess Medical Center between 2008 and 2019.

Discharge summaries were obtained from the **discharge.csv.gz** table within the MIMIC-IV-Note module. These summaries provide a comprehensive narrative of a patient's hospital course and serve as the standard reference for clinical coding. Prediction labels were derived from the **diagnoses_icd.csv.gz** table in the hospital module, which contains both ICD-9 and ICD-10 codes. Only ICD-10 codes were retained for consistency with modern clinical practice, and the label space was restricted to the top 50 most frequent codes to define a tractable prediction task. All data are fully de-identified in accordance with HIPAA Safe Harbor standards.

Table 2 summarizes the final modeling dataset after preprocessing and label filtering. Figure 1 shows the frequency distribution of the top 50 ICD codes retained for modeling.

Table 2. Final modeling dataset summary (MIMIC-IV, ICD-10, top-50 codes).

Characteristic	Value
Admissions (after label filtering)	115103
ICD-10 label space	50 codes
Training set proportion	70%
Validation set proportion	10%
Test set proportion	20%
Max note length (characters)	58596
Random seed	42

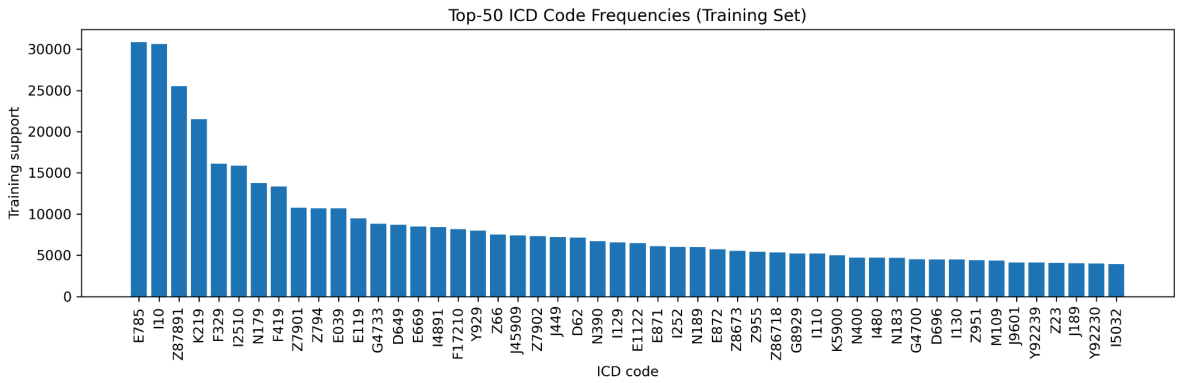

Figure 1. Frequency distribution of the top 50 ICD-10 codes in the MIMIC-IV dataset. The most frequent codes (E785, I10, Z87891) each appear in over 40,000 admissions, while the rarest codes in the top-50 set appear fewer than 5,000 times, illustrating the within-label-set imbalance that challenges all modeling approaches.

5.2 Methods

5.2.1 Problem Formulation

Automated medical coding is formulated as a multi-label text classification problem. Each hospital admission is represented by a single discharge summary, and the objective is to predict the subset of ICD-10 codes associated with that admission. Unlike single-label classification, multiple diagnoses may co-occur within a single encounter. Each admission is therefore encoded as a binary vector of length 50, where each dimension corresponds to one ICD code and the value indicates presence (1) or absence (0).

5.2.2 Data Preprocessing

Discharge summaries underwent minimal preprocessing to preserve clinically relevant terminology, abbreviations, and contextual information. Steps included: removal of null

or empty records; lowercasing for consistency; and retention of all other original formatting to avoid discarding clinical signals. Diagnosis codes were aggregated at the admission level using **hadm_id**, producing a multi-label representation for each encounter. Admissions without any top-50 labels after filtering were excluded. Multi-label binarization was performed using scikit-learn's **MultiLabelBinarizer**.

5.2.3 Baseline Model: TF-IDF with Logistic Regression

Discharge summaries were vectorized using Term Frequency–Inverse Document Frequency (TF-IDF) with both unigrams and bigrams (max_features=20,000; min_df=5; max_df=0.9). A One-vs-Rest Logistic Regression classifier (max_iter=2,000; solver=liblinear) was trained with a separate binary classifier for each of the 50 ICD codes. Logistic Regression was selected for its strong performance in high-dimensional sparse text settings and its role as a standard benchmark in clinical NLP tasks. Critically, TF-IDF vectorizes the full discharge summary without any length truncation.

5.2.4 Transformer-Based Model: BioClinicalBERT

To capture contextual relationships within clinical text, **BioClinicalBERT** (emilyalsentzer/Bio_ClinicalBERT) was fine-tuned for multi-label classification. This model is pre-trained on clinical narratives from MIMIC-III, enabling improved understanding of medical terminology and shorthand compared to general-domain language models (Alsentzer et al., 2019).

Because discharge summaries frequently exceed the 512-token architectural limit of BERT-family models, input sequences were truncated to 512 tokens, corresponding to approximately the first 1,500–2,000 characters of each note. A classification head with sigmoid activation was appended to produce independent probability estimates for each of the 50 codes. The model was trained using binary cross-entropy loss with class-specific positive weights (computed from training label frequencies) to mitigate label imbalance. Training used one epoch, a learning rate of 2e-5, weight decay of 0.01, and a batch size of 4 with gradient accumulation.

5.2.5 Threshold Optimization

In multi-label classification, a probability threshold must be chosen to convert sigmoid outputs into binary predictions. The default threshold of 0.5 is rarely optimal. Following best practices (Shamatrin, 2025), a threshold sweep was conducted over the range 0.10 to 0.90 in increments of 0.05 using the validation set. The threshold maximizing validation Micro-F1 was then applied to the held-out test set for final reporting.

5.2.6 Evaluation Metrics

Model performance was evaluated using metrics appropriate for multi-label classification (Grandini et al., 2020):

- **Micro-averaged F1 (primary metric):** aggregates true positives, false positives, and false negatives across all labels before computing F1; reflects overall performance weighted by label frequency.
- **Macro-averaged F1:** computes F1 per label and averages unweighted; ensures performance on rare codes is not obscured.

- **Precision@5 (P@5):** fraction of the model's top-5 predicted codes that are correct; simulates an assistive coding tool recommending five candidates.
- **Recall@5 (R@5):** fraction of true codes captured in the model's top-5 predictions; reflects coverage in an assistive workflow.

5.2.7 Implementation Details

All experiments were conducted in Python using Jupyter Notebooks. Data preprocessing and baseline modeling used **pandas** and **scikit-learn**. The transformer model was developed using the Hugging Face **Transformers** library (version ≥4.30) with PyTorch as the backend. Tokenization used the BioClinicalBERT fast tokenizer. Dataset splits were saved as reproducible CSV files indexed by **hadm_id** with a fixed random seed of 42.

6. Results

Both models were evaluated on the same held-out test set. Table 3 reports the primary performance metrics. Figure 2 visualizes the baseline results. Figure 3 provides a direct comparison of all four metrics across both models.

Table 3. Model performance on the held-out test set (top-50 ICD-10 codes, MIMIC-IV).

Metric	Baseline (TF-IDF + LR)	Transformer (BioClinicalBERT)	Best threshold	Delta
Micro-F1 (primary)	**0.5303**	0.4327 (threshold = 0.65)	0.65	−0.098
Macro-F1	**0.4179**	0.4081 (threshold = 0.65)	0.65	−0.010
Precision@5	**0.5695**	0.3998	N/A	−0.170
Recall@5	**0.6123**	0.4275	N/A	−0.185

Note: Transformer F1 values are reported at the optimal threshold (0.65) derived from the validation-set threshold sweep. At the default threshold of 0.50, Micro-F1 = 0.3917 and Macro-F1 = 0.3778, which are even further below the baseline.

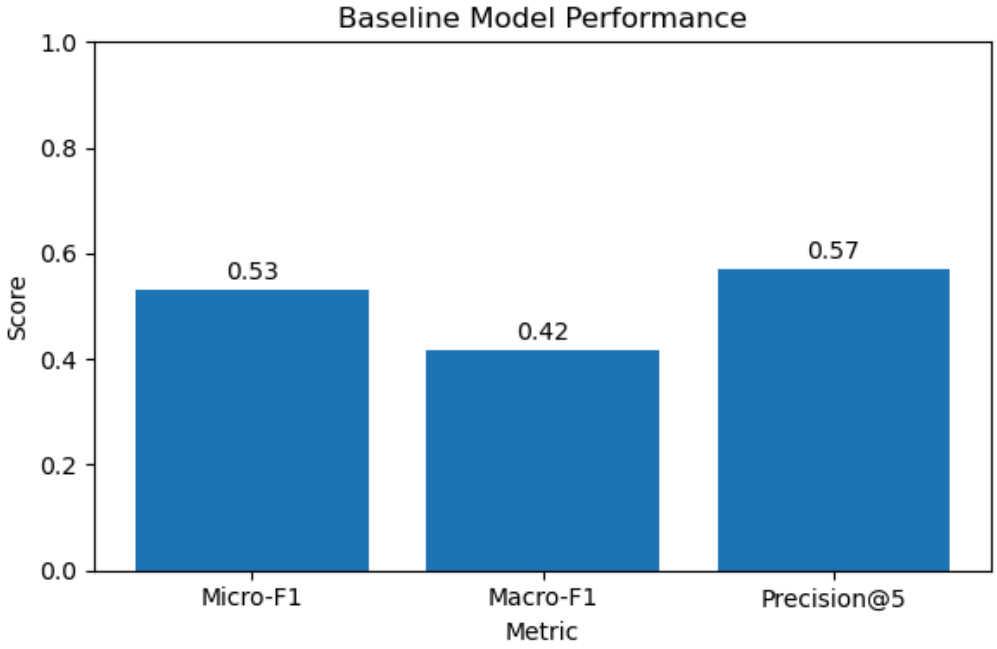

Figure 2. Performance of the TF-IDF + Logistic Regression baseline on the held-out test set. Precision@5 (0.57) is the highest metric, indicating reasonable assistive coding potential in a top-5 recommendation workflow.

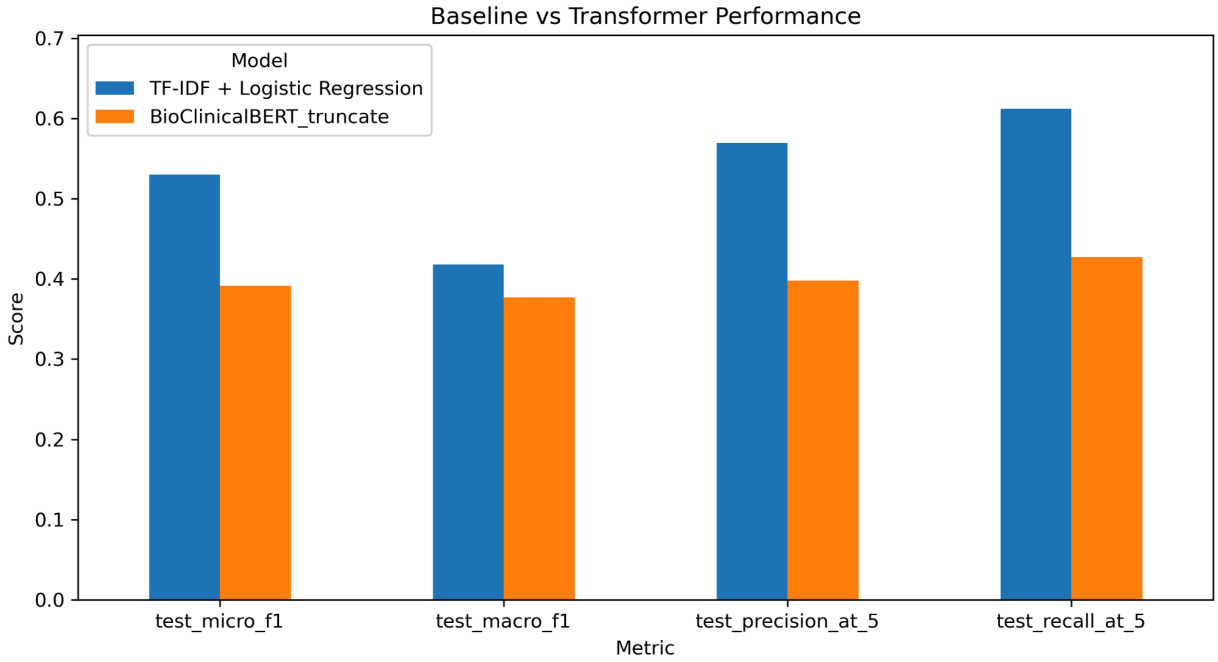

Figure 3. Direct metric comparison between the TF-IDF baseline and BioClinicalBERT transformer. The baseline outperforms the transformer on all four metrics, with the largest gap observed in Precision@5 and Recall@5 (+0.17 and +0.18, respectively).

6.1 The Baseline Outperforms the Transformer

Contrary to the assumption that domain-specific transformer models will automatically improve on TF-IDF baselines, the TF-IDF + Logistic Regression model outperformed BioClinicalBERT on every evaluation metric. The baseline achieved a Micro-F1 of 0.5303 compared to the transformer's best of 0.4327, a Macro-F1 of 0.4179 vs. 0.4081, and Precision@5 of 0.5695 vs. 0.3998. The performance gap is largest for P@5 and R@5, which are the most clinically relevant metrics in an assistive coding context.

This result is consistent with findings from Edin et al. (2023), who systematically demonstrated that truncation of discharge summaries at 512 tokens is the primary bottleneck for transformer models on MIMIC-IV. Discharge summaries in this dataset average several thousand words; the 512-token limit captures only the opening sections, which tend to contain admission complaints and history but not the full diagnostic summary typically placed at the end of the document. The TF-IDF model, by contrast, operates over the full note and is not subject to this constraint.

6.2 Effect of Threshold Selection

The default sigmoid threshold of 0.50 produced a Micro-F1 of 0.3917 for the transformer. The validation-set threshold sweep revealed that performance improves substantially with a higher threshold: Micro-F1 peaks at 0.4327 (threshold = 0.65) and Macro-F1 at 0.4081 (also threshold = 0.65). This indicates that the model is over-predicting at lower thresholds — likely a consequence of class-imbalance patterns in training — and that conservative thresholding substantially recovers discriminative power. Figure 4 shows the full threshold sweep curve.

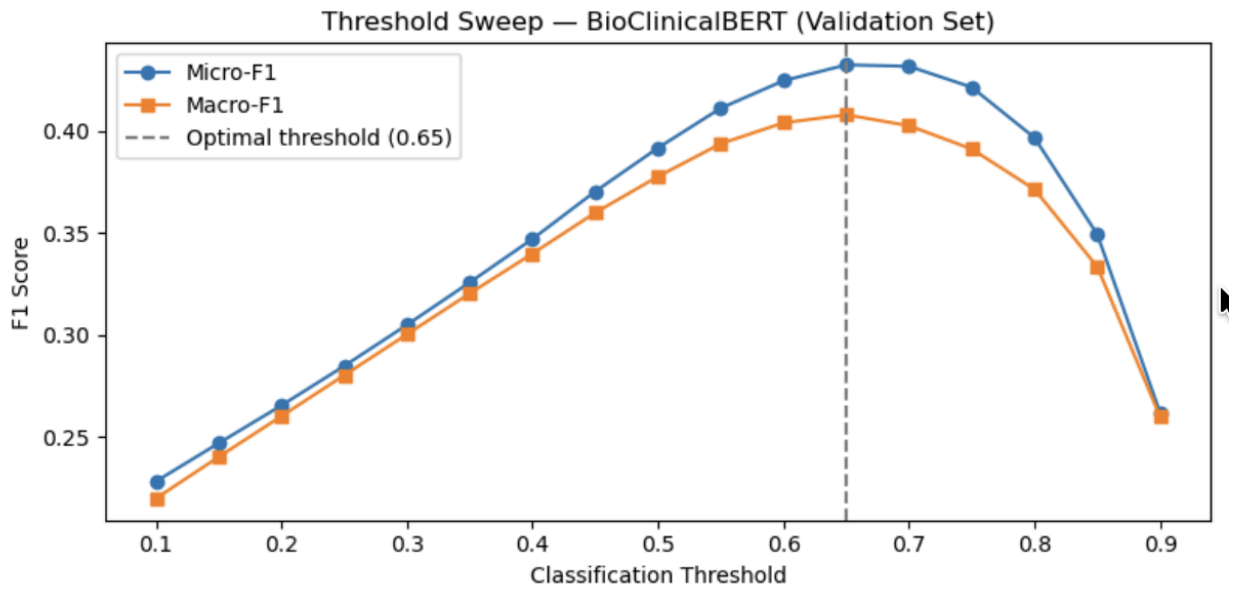

Figure 4. Validation-set threshold sweep for BioClinicalBERT (Micro-F1 and Macro-F1 vs. classification threshold). Both metrics peak near threshold = 0.65, well above the commonly used default of 0.50, demonstrating the importance of threshold optimization in multi-label clinical NLP tasks.

It should be noted that Precision@5 and Recall@5 are invariant to threshold selection by design, as they are rank-based metrics computed over the top-5 probability scores rather than over a binary cutoff. The constant P@5 and R@5 values observed across all threshold values in the sweep are therefore expected behavior.

6.3 Impact of Label Frequency (Long-Tail Distribution)

Per-label analysis confirmed that model accuracy is strongly correlated with ICD code frequency in the training set for both models. High-frequency codes — such as I10 (essential hypertension, ~43,000 occurrences) and E785 (hyperlipidemia, ~44,000 occurrences) — achieved substantially higher individual F1 scores than lower-frequency codes in the top-50 set. This reflects the well-known long-tail distribution in clinical coding datasets, where even the top-50 label set spans nearly an order of magnitude in frequency.

The baseline model showed a sharper decline in per-label performance for low-frequency codes, while the transformer showed somewhat flatter degradation across the label range — likely because the pre-trained contextual embeddings provide more generalizable representations for rare conditions than sparse TF-IDF features. However, both models struggle meaningfully with the lowest-frequency labels in the top-50 set.

6.4 Effect of Clinical Note Length

For the TF-IDF model, discharge summaries are treated as unordered bags of terms, meaning length does not directly constrain the model's access to information. However, very long notes may dilute important signals due to feature sparsity across a 20,000-term vocabulary. Error analysis using quintile-binned Jaccard similarity showed modest performance degradation for the longest notes in the test set, consistent with signal dilution.

For the transformer, the 512-token truncation limit imposes a hard constraint that is particularly costly for long notes. Error analysis suggests that many false negatives — true codes that the model fails to predict — involve diagnoses documented in the latter sections of the note, beyond the truncated window. This is the primary structural explanation for the performance gap between the two models.

6.5 Error Analysis

Qualitative error analysis identified several systematic patterns shared by both models:

- **False negatives — implicit diagnosis:** diagnoses implied indirectly rather than stated explicitly (e.g., clinical management patterns suggesting a diagnosis not yet named in the note).
- **False negatives — distributed evidence:** clinical evidence distributed across multiple sections of the summary, requiring the reader to synthesize information from different parts of the document.

- **False negatives — truncation (transformer only):** important diagnostic summaries appearing beyond the 512-token window.
- **False positives — historical conditions:** past medical history conditions (coded with Z-codes) sometimes misclassified as active diagnoses.
- **False positives — negated conditions:** negated clinical phrases (e.g., "no evidence of pneumonia," "ruled out for sepsis") misinterpreted as positive findings by both models.
- **False positives — overgeneralization:** common clinical terms that co-occur with many diagnoses leading to spurious positive predictions.

These findings highlight the inherent difficulty of interpreting clinical narratives, where negation, temporality, and contextual qualification play critical roles that neither TF-IDF nor truncated-BERT fully captures.

6.6 Practical Implications for Assistive Coding

Despite the overall dominance of the baseline, both models show meaningful potential in an assistive coding workflow. The baseline's Precision@5 of 0.57 and Recall@5 of 0.61 mean that, on average, when five candidate codes are surfaced to a human coder, more than half are correct and approximately 61% of the true codes are captured in that shortlist. This substantially reduces the cognitive search space for the coder compared to manually reviewing 70,000+ codes.

The transformer's R@5 of 0.43, while lower than the baseline, still represents meaningful coverage in a recommendation context — roughly 43 out of every 100 true codes would appear in the top-5 suggestion list. The transformer may be more useful in settings where the model is used to surface unusual codes for secondary review, given its potentially better generalization across the label distribution.

6.7 Summary of Results

In summary, the TF-IDF + Logistic Regression baseline outperformed the fine-tuned BioClinicalBERT model on all four evaluation metrics on this dataset. The transformer's underperformance is primarily attributable to aggressive input truncation at 512 tokens, which discards clinically critical information from the latter portions of discharge summaries. Threshold optimization via validation-set sweep partially recovers transformer performance (Micro-F1 improves from 0.39 to 0.43), but does not close the gap to the baseline. These results illustrate that document length is a decisive factor in clinical NLP model selection, and that computationally simpler methods can outperform deep learning models when architectural constraints conflict with input characteristics.

7. Discussion

7.1 Key Findings

This study yields three primary findings. First, a well-configured TF-IDF + Logistic Regression model provides a strong and hard-to-beat baseline for top-50 ICD coding from MIMIC-IV discharge summaries, achieving Micro-F1 = 0.53 and Precision@5 = 0.57. This is consistent with the literature showing that bag-of-words methods remain competitive on clinical NLP tasks when document length penalizes transformer competitors (Edin et al., 2023; Nielsen & Kılıç, 2025).

Second, the BioClinicalBERT transformer, while architecturally sophisticated and pre-trained on relevant clinical text, underperforms the baseline when limited to 512-token input truncation. This reinforces the core finding of Edin et al. (2023): for MIMIC-IV discharge summaries, truncation is not a minor limitation but a fundamental structural problem that removes the most clinically informative text from the model's input.

Third, classification threshold optimization is essential for multi-label transformer models. The default threshold of 0.50 yields Micro-F1 = 0.39, substantially below the optimized threshold of 0.65 which yields 0.43 — a relative improvement of approximately 10%. This finding generalizes to any multi-label clinical NLP system using sigmoid outputs and should be treated as a mandatory post-training calibration step.

7.2 Limitations

- **Input truncation:** The 512-token limit imposed by the BioClinicalBERT architecture discards an estimated majority of each discharge summary. Future work should address this with sliding-window, hierarchical, or long-document transformer architectures (e.g., Longformer, BigBird, PLM-ICD).
- **Single training epoch:** The transformer was trained for only one epoch due to computational constraints. Additional training epochs may substantially improve performance and are a necessary next step before drawing strong comparative conclusions about architecture quality.
- **Label scope:** By focusing only on the top 50 codes, this study does not address the extreme multi-label challenge of the full 70,000+ ICD-10 code space. The dynamics of rare-code prediction are substantially different and would require different modeling strategies.
- **Population bias:** The MIMIC dataset is sourced exclusively from a single tertiary care ICU. Findings may not generalize to outpatient settings, community hospitals, or health systems with different documentation practices.

- **Retrospective design:** This is a retrospective study using historical EHR data. It cannot capture dynamic human-AI interaction effects that would appear in a live coding assistance deployment.
- **P@K and R@K invariance:** The rank-based evaluation metrics (Precision@5, Recall@5) are constant across classification threshold values, which limits their utility for threshold selection. Label-specific threshold tuning and alternative ranking metrics (e.g., AUC-ROC per label) should be explored in future work.

7.3 Recommendations

Based on these findings, the following recommendations are offered for researchers and practitioners building automated ICD coding systems:

- **Address truncation before choosing a model architecture:** For discharge summaries, no transformer model should be evaluated without first addressing the long-document problem. Sliding-window pooling, hierarchical encoding, or direct fine-tuning of long-document models (Longformer, PLM-ICD) are the most promising approaches.
- **Always optimize classification threshold on a held-out validation set:** The default 0.50 threshold is rarely optimal for imbalanced multi-label problems. A sweep over 0.10–0.90 at 0.05 increments adds minimal computation and yields meaningful gains.
- **Use P@K and R@K as primary clinical utility metrics:** In a clinical workflow where the system recommends a shortlist of codes to a human coder, Precision@K and Recall@K more directly measure real-world value than threshold-dependent F1 scores.
- **Benchmark TF-IDF seriously:** TF-IDF baselines are often dismissed as too simple in deep learning literature. For long-document tasks with architectural constraints, they should be treated as a strong competitive baseline and reported alongside neural methods.

7.4 Future Work

Several natural extensions of this work are identified:

- Fine-tune a long-document transformer (Longformer or PLM-ICD) that can process full discharge summaries without truncation, which is the most likely path to substantially outperforming the TF-IDF baseline.
- Train the transformer for additional epochs (minimum 3–5) to allow the model to more fully adapt its pre-trained weights to the ICD prediction task.
- Explore label-specific threshold tuning, where each of the 50 ICD codes receives an individually optimized threshold calibrated on the validation set.
- Expand the label space beyond the top 50 codes to evaluate model performance across the long tail of ICD-10, where rare-code prediction is the primary clinical challenge.

- Investigate agentic approaches such as CLH and MedDCR for open-set coding across the full ICD-10 space, building on the literature reviewed in Section 3.4.
- Conduct prospective evaluation in a live clinical coding workflow to measure the actual impact on coder productivity and claim accuracy.

8. Conclusion

This paper presented and evaluated two approaches to automated multi-label ICD-10 coding from clinical discharge summaries using the MIMIC-IV dataset: a TF-IDF + Logistic Regression baseline and a fine-tuned BioClinicalBERT transformer. Against expectations, the baseline model outperformed the transformer on all four evaluation metrics, achieving Micro-F1 = 0.53 vs. 0.43 (at optimal threshold), Macro-F1 = 0.42 vs. 0.41, Precision@5 = 0.57 vs. 0.40, and Recall@5 = 0.61 vs. 0.43.

The primary explanation for this result is the 512-token truncation imposed by the transformer architecture, which discards the majority of long discharge summaries and removes diagnostically critical content. This finding replicates and extends the results of Edin et al. (2023), who identified truncation as the principal bottleneck for BERT-family models on MIMIC. The threshold sweep analysis further demonstrated that default probability thresholds substantially underestimate transformer performance and that calibration is essential.

From a clinical standpoint, both models show promise as assistive coding tools. A Precision@5 exceeding 0.57 means that a shortlist of five candidate codes would contain a majority of correct codes, reducing the search burden on human coders and improving workflow efficiency. Closing the performance gap between the models, and between both models and expert coders, will require addressing document length either through architectural innovation or through the emerging agentic workflow paradigms that mirror the multi-step reasoning of professional medical coders.

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